Embedded Systems Tutorial 2 Solutions

1) a) 1024

b) $f_{timer} = \frac{f_{clk}}{prescaler} = \frac{8 \times 10^6}{1024}$

$\Delta t = \frac{1}{f_{timer}} = \frac{prescaler}{f_{clk}} = \frac{1024}{8 \times 10^6}$

$\Delta t = 1.28 \times 10^{-4}$ s or 128 μs or 0.128 ms

c) $T = \Delta t \times 2^N$

$T = \frac{1024}{8 \times 10^6} \times 2^{16}$

$T = 8.39$ s

d) 10

e)
```
#define HUEY 10
#define DEWEY 400
#define LOUIE 10
```
line 26 flasher(HUEY, DEWEY, LOUIE);

f) $Duty\text{-}cycle = \frac{ON_TIME}{ON_TIME + OFF_TIME}$

This will vary!

$Duty\text{-}cycle = \frac{HUEY}{DEWEY - LOUIE + HUEY}$

Initial Duty-cycle $= \frac{10}{400 - 9 + 10} = \frac{10}{401} = 2.49\%$

Final Duty-cycle $= \frac{10}{400 - 0 + 10} = \frac{10}{410} = \frac{1}{41}$

$= 2.44\%$

Question 2

```
1  int main(void)
2  {
3      unit32_t button_tick=0, display_tick=0;
4
5      init_game_parameters();
6
7      pacer_init(GAME_RATE);
8
9      while(1)
10     {
11         pacer_wait();
12
13         button_tick++;
14         display_tick++;
15
16         update_game();
17
18         if (button_tick >= (GAME_RATE/BUTTON_RATE))
19         {
20             get_button();
21             button_tick=0;
22         }
23
24         if (display_tick >= (GAME_RATE/DISPLAY_RATE))
25         {
26             make_display();
27             display_tick=0;
28         }
29     }
30
31     return 0;
32
33 }
```